

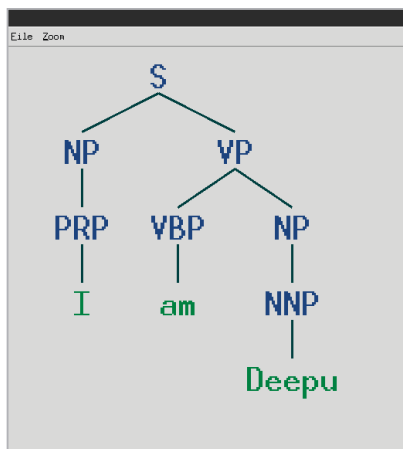


Figure 5: An example for a parse tree

can be combined with Hadoop to provide it with the power of AI and machine learning. PySpark offers distributed computing and APIs for Big Data processing, while Hadoop provides storage, scalability, and fault-tolerance for handling large data sets. This combination with Hadoop makes PySpark an important tool in data science and Big Data analytics.

Now, let us install PySpark in our system. As always, Anaconda Navigator can be used to install PySpark. However, if you execute the command `'pip3 install pyspark[sql, ml, mllib]'` in the terminal, all three important PySpark modules will be installed instantly. You are free to use either of these methods to install PySpark. After installation, execute the command `'pyspark --master local[*]'` in the terminal to launch a PySpark interactive shell with the local machine as the master, while using all available cores for parallel processing. Figure 6 shows the PySpark interactive shell.

Let us now consider the simple Python script named *spark.py* shown below, which uses PySpark to read a CSV file from our system and display the content on the terminal (line numbers are added for ease of explanation).

```
1. from pyspark.sql import
```

```

Welcome to
      _ _ _ _ _
     / /   / /
    / /   / /
   / /   / /
  / /   / /
 / /   / /
/_/_/_/_/_/ version 3.3.1

Using Python version 3.9.12 (main, Apr 5 2022 06:56:58)
Spark context Web UI available at http://10.7.45.41:4040
Spark context available as 'sc' (master = local[*], app id = local-1676324644625).
SparkSession available as 'spark'.
>>>
  
```

Figure 6: The PySpark interactive shell

```

+-----+-----+
|      Name|Centuries|
+-----+-----+
|    Sachin|      49|
|    Kohli |      43|
|  Ponting |      30|
|    Rohit |      29|
|Jayasuriya|      28|
+-----+-----+
only showing top 5 rows
  
```

Figure 7: Output of the Python script *spark.py*

```

SparkSession
2. spark = SparkSession.builder.
  appName("ReadCSV").getOrCreate()
3. df = spark.read.csv("cricket.csv",
  header=True, inferSchema=True)
4. df.show(5)
  
```

First, let us try to understand the working of the Python script *spark.py*. Line 1 imports the *SparkSession* class from the *pyspark.sql* module in the PySpark library. The *SparkSession* class provides methods for manipulating data with PySpark, and is used to create data frames and execute SQL queries in a distributed computing environment. Line 2 creates a *SparkSession* object named *spark*. The *getOrCreate()* method is used to either get the existing *SparkSession* or create a new one if it does not

exist. Line 3 reads a CSV file named *cricket.csv*. Finally, Line 4 prints the first five rows of the CSV file *cricket.csv*. Figure 7 shows the output of the Python script *spark.py*.

Now let us address a crucial question: We have previously performed the same tasks using Pandas quite some time ago. In that case, what is the need for PySpark? While Pandas is a great tool for handling and analysing data from small to medium-sized data sets, PySpark is a scalable distributed computing platform designed to manage large data volumes. If your data set is large, PySpark is the better choice. Pandas is only suitable for small to medium-sized data sets that do not require a distributed system.

Now, it is time to wind up this article. Here we first delved deep into the theory of neural networks and then turned our attention to NLP. We utilised PyTorch and NLTK to gain an understanding of NLP. Afterwards, we introduced PySpark and thus marked our first visit into the fields of data science and Big Data analytics in this series. In the upcoming article in this series, we will focus on computer vision using AI and machine learning, and reacquaint ourselves with OpenCV, a library for real-time computer vision. **END** 

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